

## Spring 2026A ASTRONOMY 302 Observing Proposal

**Abstract of Scientific Justification (300 words max) (2 points):** Summarize your program. Why is this observing program scientifically interesting? What target(s) are you trying to observe and with what observational modes? What science will your program uncover?

Our program seeks to measure the magnitude of two classical cepheids, J013401.23+303113.5 and J013352.47+303844.2, in the M33 Galaxy using the 90" Bok Telescope on Kitt Peak. Cepheids, due to extensive past research and their properties, are one of the most reliable 'cosmic rulers' or standard candles. Specifically, there is a simplified period-luminosity relationship which estimates the distance to the cepheid as well as a more accurate and complete model for the variations in brightness versus time. This variation is periodic in nature, and thus we can take multiple spaced out observations to get a constraint on the Cepheid's physical properties. We will compare and add our observations in the g and the i bands with observations done previously by one of Dr. Milne's students who has mapped multiple M33 cepheids' magnitudes over time. We will finally fit the model to our combined data in order to get a more accurate distance measurement which we will report. Thus, our overall goal is to use our measurements to further constrain a distance to the M33 Galaxy and contribute more towards science related to galaxy distances, such as the Hubble tension.

**Telescope Requested (0.5 points): Bok 90"**

**Instrument (0.5 points): 90 Prime**

**Tolerable Moon Brightness (dark, grey, bright, etc) (0.5 points):** Dark/very low moon brightness is the limit for our observing due to our faint targets. Due to the moon's phase being in the gibbous phase for the allotted potential dates of our observation, we require that the moon has not risen above the horizon yet. This leaves October 2nd and 3rd to conduct our observations in full.

### Target List (2 points)

| Object Name         | Right Ascension        | Declination            | V magnitude | Object Type       | Visibility      |
|---------------------|------------------------|------------------------|-------------|-------------------|-----------------|
| M33                 | 01 33<br>50.8965749232 | +30 39<br>36.630403128 | 5.72        | Galaxy            | Visible May-Jan |
| J013401.23+303113.5 | 01 34<br>01.1961715224 | +30 31<br>13.402671456 | 19.1        | Classical Cepheid | Visible May-Jan |
| J013352.47+303844.2 | 01 33<br>52.4402010666 | +30 38<br>44.082599552 | 19.4        | Classical Cepheid | Visible May-Jan |

### Experimental Design & Technical Description (4.5 points total)

How much time is required for your program and what signal-to-noise is necessary for your observations? (2 points)

We aim for a SNR of 100. To achieve a SNR of 100, we require an exposure time of 30 minutes, however, we will be using both the g and i filters for two separate exposures, totalling to 1 hour of observation time. Due to the g band being more impacted by moonlight, we will conduct our g band observations first.

How will these observations contribute toward the accomplishment of the goals outlined in the abstract? **(2 points)**

The observations we make will add more measurements to the Cepheid's observed magnitude over time, and via model fitting, we will be able to further constrain physical quantities of the cepheid, including its distance. Knowing the distance to the Cepheids will give us a better distance estimate to M33, which will accomplish our program's goal.

Back Up Plan. Describe a potential project that can be done with poor weather or archival data. **(0.5 points)**

Given poor weather for our observations, we can take Dr. Milne's unreduced data of the same M33 cepheids and fit the data in a similar manner, and check if the derived values match with the accepted values for distance to M33.